DEWDRIVE · DESIGN DECISION RECORD

DewDrive TRL 3 recommendations accepted

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DOCUMENT CONTROL

Document ID	DWD-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/decisions/0002-recommendations-accepted.md @ 4f86e9e		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1, O4 and O5 decided by Amish on 2026-10-02 as recommended in DWD-DEC-001
0.2	2026-09-26	Amish Chadha	Budget top-up approved by Amish (O3 decided)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: TRL 3 recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items D9 to D14 and, on 2026-09-26, the budget top-up (D15, closing O3); items O1, O4 and O5 were decided by Amish on 2026-10-02 as recommended in the design decisions register (DWD-DEC-001, open items 3 to 5): "i approve your recommendations for all 555 open decisions."

Context

The TRL 3 review note (`docs/REVIEW.md`, session 2026-09-25, TRL 3) listed six new items with a recommendation, raised by DWD-CAL-001 v0.1 after DWD-DDR-001 was written. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of it. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so nothing here is built, bought or tested.

Decision

Table 1. Decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation".

#	Item (REVIEW.md, TRL 3)	Decision	What changed in the repo
D9	Night airflow through the bed (item 2)	Option (a): seal the tray edges and draw the night air down through the mesh-floored trays	`model.py` : 1 mm sealing baffle between trays and walls, trays on an EPDM edge gasket; south inlet flap moved above the trays (800 x 40 mm, was 800 x 70 mm below them); outlet and fan stay below the trays. DWD-CAL-001 v0.2 now models through-flow as the design of record. BOM line 4 $10.00 to $11.50 each
D10	Salt loading (item 3; amends D3)	Lower the $CaCl_2$ to 25 wt %: 1.0 kg in 3.0 kg of gel (was 1.3 kg in 2.7 kg, 32.5 wt %)	Bed depth 8.3 to 9.3 mm in `model.py` ; BOM line 5 $28 to $29; pores now fill at 71 % RH, not 51 %
D11	Humid-night protection (item 4)	Adopt both: a drip tray under each sorbent tray, and a logger rule that stops the fan above 70 % RH	New BOM line 15 (four drip screens, $20): two staggered layers of 20 x 6 mm aluminium channels running down the slope, painted black, with a closed brine sump at the low end (0.58 L in all). They catch drips without blocking the air and vapour path. The fan rule is stated in DWD-PRC-001; firmware beyond a sketch is TRL 4 and on hold
D12	Bed-to-condenser radiation screen (item 5)	Evaluate on paper before any build	Done in DWD-CAL-001 v0.2 (C5, C10, D1): low-emissivity screens would raise the yield to 0.82 L per day but push the dry bed to 166 °C at stagnation, beyond the usual polycarbonate rating. Not adopted; the drip screens are painted black instead. A follow-up is proposed below (O4)
D13	Wind restraint (item 6)	Two auger ground anchors are the default; about 50 kg of ballast is the alternative	Wording in DWD-PRC-001, DWD-REQ-001 (R10) and the drawing notes; anchors were already in BOM line 1
D14	Budget margin (item 7)	Keep `budget_usd` at $500	`project.yaml` unchanged at 500. The premise that items 2 to 4 add "only a few dollars" did not hold: the BOM is now $515 (see O3, closed by D15)
D15	Budget top-up (O3)	Budget top-up to $520: decided by Amish, 2026-09-26 ("I am ok with the budget top ups")	`project.yaml` `budget_usd` 500 to 520; R11 in DWD-REQ-001 v0.5 now $520 or less and met at $515; DWD-CAL-001 v0.3 re-run

Results of the decided design (DWD-CAL-001 v0.2), before and after:

Quantity	Before (CAL-001 v0.1, as drawn)	After (v0.2, design of record)
Water at 40 % night RH (R1, 0.5 L)	0.20 L per day	0.57 L per day
Water at 25 % night RH (R2, 0.25 L)	0.09 L per day	0.45 L per day
Pore-filling humidity (R7)	51 % RH	71 % RH
Pores filled by one 90 % RH night	46 %	43 %; 50 % after three such nights with the fan rule
Bed at noon; stagnation	104 °C; 105 °C	112 °C; 122 °C
Condenser rise (R5, 15 K)	13.1 K	12.1 K
Solar-to-water efficiency	2.5 %	7.1 %
Mass; box	51.4 kg; 30.9 kg	53.9 kg; 33.4 kg
Parts cost (R11, $500)	$488	$515

Items left open by this record

Table 2. Items left open by this record, all since decided.

#	Item	Status
O1	First partner and region for co-design, which also supplies the site climate data	**Decided by Amish, 2026-10-02:** first candidate type to approach (not yet agreed), a university dryland field station in a hot semi-arid region with night humidity of about 25 to 40 % and an existing weather record, with a water and sanitation NGO for the later community trial (DWD-DEC-001, item 5)
O3	Budget: the BOM is $515 against $500 (R11 not met)	**Decided by Amish, 2026-09-26: budget top-up to $520 (option a); see D15.** Original options: Options: (a) raise `budget_usd` to $520; (b) cut the drip screens to a drip gutter under the low edge of each tray (about $8 in all, weaker protection); (c) evaluate a lighter condenser on paper, since the black screens cut its peak load from 333 W to 299 W. Recommendation: (c) first, then (a) if it does not close the gap
O4	Low-emissivity screen with overheat protection	New. The low-e screen gives 0.82 L per day but a 166 °C stagnation bed; it would need a stagnation vent or a glazing rated well above that. Recommendation: evaluate on paper at TRL 3. **Decided by Amish, 2026-10-02:** one short paper study of a passive stagnation vent; if it cannot hold a dry bed near 122 °C at a modest cost, drop the low-emissivity screens (DWD-DEC-001, item 4)
O5	Glazing temperature rating	New. With black drip screens the dry bed stagnates at about 122 °C, near the usual 120 °C service rating of twin-wall polycarbonate. Recommendation: specify a sheet rated 130 °C or more, or confirm from supplier data that the inner skin, which runs cooler than the bed, stays inside the rating. **Decided by Amish, 2026-10-02:** a sheet rated 130 °C or more; a 120 °C sheet only if the supplier's data and the inner-skin temperature measured at the TRL 4 stagnation test both show it stays inside its rating (DWD-DEC-001, item 3)

Consequences

- DWD-PRB-001 v0.4, DWD-PRC-001 v0.4, DWD-REQ-001 v0.4 and DWD-CAL-001 v0.2 carry the decided design. R7 is restated to cover the tray and drip-screen assembly.

- Drawing DWD-DWG-001 moves to Rev P2; STEP, STL and all media are regenerated from `model.py`.
- Requirement status: 10 met, 1 not met (R11, cost), 2 not verifiable at TRL 3 (R6, R10). After the budget top-up to $520 (D15, 2026-09-26), R11 is met: 11 met, 2 not verifiable at TRL 3.
- `trl: 3` and `trl_target: 3` are unchanged. The drip-screen sump test, measured isotherms and the fan rule firmware are TRL 4 work and on hold.